

Pizza Sales and Delivery Performance Analysis

Project Overview

The fast-paced food delivery industry relies heavily on operational efficiency and customer satisfaction. This project involves a comprehensive analysis of pizza delivery data to explore key performance metrics, popular choices, and operational bottlenecks. With rising consumer expectations and increased traffic challenges, it is crucial to assess how various factors—such as traffic, time of day, and pizza preferences—influence delivery performance and customer behavior.

Objective

This project analyzes delivery and order data for a pizza business to:

- Identify peak sales periods and best-performing pizza types.
- Evaluate delivery performance and delays.
- Understand how external factors like traffic and peak hours affect operations.
- Discover customer preferences regarding size, toppings, and payment methods.
- Recommend actions to optimize delivery efficiency and improve customer satisfaction.

Why This Project?

- Pizza delivery is a competitive market where speed, product variety, and customer experience are paramount.
- Delays can negatively impact brand perception, and understanding ordering behavior helps target promotions and operational improvements.
- This project helps uncover performance bottlenecks and customer trends to inform better business decisions

Key Queries and Insights :

1.Order Performance Overview:

Input:

```
SELECT * FROM pizza_sales_db.pizza_sales;
```

Output:

	Order ID	Restaurant Name	Location	Order Time	Delivery Time	Delivery Duration (min)	Pizza Size	Pizza Type	Toppings Count	Distance (km)	
▶	ORD001	Domino's	New York, NY	05-01-2024 18:30	05-01-2024 18:45	15	Medium	Veg	3	2.5	▶
	ORD002	Papa John's	Los Angeles, CA	14-02-2024 20:00	14-02-2024 20:25	25	Large	Non-Veg	4	5	▶
	ORD003	Little Caesars	Chicago, IL	21-03-2024 12:15	21-03-2024 12:35	20	Small	Vegan	2	3	L
	ORD004	Pizza Hut	Miami, FL	10-04-2024 19:45	10-04-2024 20:10	25	XL	Cheese Burst	5	4.5	▶
	ORD005	Marco's Pizza	Dallas, TX	05-05-2024 13:00	05-05-2024 13:20	20	Medium	Non-Veg	3	2	▶
	ORD006	Domino's	New York, NY	06-01-2024 19:00	06-01-2024 19:20	20	Medium	Non-Veg	3	3	▶
	ORD007	Papa John's	Los Angeles, CA	07-01-2024 20:15	07-01-2024 20:40	25	Large	Veg	4	4.5	▶
	ORD008	Little Caesars	Chicago, IL	08-01-2024 18:30	08-01-2024 18:50	20	Small	Vegan	2	2	L
	ORD009	Pizza Hut	Miami, FL	09-01-2024 19:45	09-01-2024 20:10	25	XL	Cheese Burst	5	5	▶
	ORD010	Marco's Pizza	Dallas, TX	10-01-2024 13:00	10-01-2024 13:20	20	Medium	Non-Veg	3	3.5	▶
	ORD011	Domino's	San Francisco, CA	11-01-2024 17:30	11-01-2024 17:50	20	Large	Veg	4	4	L
	ORD012	Papa John's	Houston, TX	12-01-2024 19:00	12-01-2024 19:25	25	Medium	Non-Veg	3	3	▶
	ORD013	Little Caesars	Phoenix, AZ	13-01-2024 18:15	13-01-2024 18:35	20	Small	Vegan	2	2.5	▶

Insight: "August records the highest number of orders, indicating it as the peak month for business activity."

2. Describe pizza sales:

Input:

```
DESCRIBE pizza_sales;
```

Output:

	Field	Type	Null	Key	Default	Extra
►	Order ID	text	YES		NULL	
	Restaurant Name	text	YES		NULL	
	Location	text	YES		NULL	
	Order Time	text	YES		NULL	
	Delivery Time	text	YES		NULL	
	Delivery Duration (min)	int	YES		NULL	
	Pizza Size	text	YES		NULL	
	Pizza Type	text	YES		NULL	
	Toppings Count	int	YES		NULL	
	Distance (km)	double	YES		NULL	
	Traffic Level	text	YES		NULL	
	Payment Method	text	YES		NULL	
	Is Peak Hour	text	YES		NULL	
	Is Weekend	text	YES		NULL	
	Topping Density	double	YES		NULL	
	Order Month	text	YES		NULL	

Insight: "The schema description confirms that the dataset contains essential order attributes including delivery duration, pizza specifications, and traffic levels, facilitating comprehensive sales analysis."

3. Total Order:

Input:

```
SELECT COUNT(`Order ID`) AS Total_Orders FROM pizza_sales;
```

Output:

	Total_Orders
▶	1004

Insight: "The total number of orders placed in the dataset is 1004, serving as a baseline for all subsequent sales performance analysis."

4. Average Delivery Time:

Input:

```
SELECT AVG(`Delivery Duration (min)`) AS Avg_Delivery_Time FROM pizza_sales;
```

Output:

	Avg_Delivery_Time
▶	29.4920

Insight: "The average delivery duration is approximately 29.49 minutes, providing a benchmark for evaluating operational efficiency and customer delivery expectations."

5. Monthly Order Distribution:

Input:

```
1 • SELECT `Order Month`, COUNT(`Order ID`) AS Total_Orders
2 FROM pizza_sales
3 GROUP BY `Order Month`
4 ORDER BY Total_Orders DESC;
```

Output:

	Order Month	Total_Orders
▶	August	117
	September	105
	March	94
	April	91
	June	91
	January	88
	February	86
	December	80
	October	75
	May	68
	November	60
	July	49

Insight: "August has the highest number of orders, highlighting it as the peak period for business."

6. Pizza Type Popularity:

Input:

```
1 • SELECT `Pizza Type`, COUNT(`Order ID`) AS Order_Count
2 FROM pizza_sales
3 GROUP BY `Pizza Type`
4 ORDER BY Order_Count DESC
5 LIMIT 5;
```

Output:

	Pizza Type	Order_Count
►	Non-Veg	216
	Veg	202
	Cheese Burst	188
	Vegan	132
	Sicilian	64

Insight: "August has the highest number of orders, highlighting it as the peak period for business."

7. Weekend vs. Weekday Orders:

Input:

```
1 • SELECT `Is Weekend`, COUNT(`Order ID`) AS Total_Orders
2 FROM pizza_sales
3 GROUP BY `Is Weekend`;
```

Output:

	Is Weekend	Total_Orders
▶	FALSE	718
	TRUE	286

Insight: "Weekdays generate nearly three times more orders than weekends, indicating higher demand during the work week."

8. Peak Hour Demand:

Input:

```
1 • SELECT `Order Hour`, COUNT(`Order ID`) AS Order_Count
2 FROM pizza_sales
3 GROUP BY `Order Hour`
4 ORDER BY Order_Count DESC;
```

Output:

	Order Hour	Order_Count
►	19	328
	18	312
	20	306
	13	43
	14	6
	17	4
	21	3
	12	2

Insight: "Order volume peaks around 7 PM, clearly signaling a high-demand dinner rush period."

9. Customer Topping Preferences:

Input:

```
1 • SELECT `Toppings Count`, COUNT(`Order ID`) AS Order_Count
2 FROM pizza_sales
3 GROUP BY `Toppings Count`
4 ORDER BY Order_Count DESC;
```

Output:

	Toppings Count	Order_Count
▶	3	319
	4	240
	5	204
	2	198
	1	43

Insight: "The majority of customers prefer 3 toppings on their pizzas, which can help in streamlining inventory."

10. Location Performance Analysis:

Input:

```
1 • SELECT Location, COUNT(`Order ID`) AS Total_Orders
2 FROM pizza_sales
3 GROUP BY Location
4 ORDER BY Total_Orders DESC;
```

Output:

	Location	Total_Orders
▶	Atlanta, GA	78
	Milwaukee, WI	71
	Louisville, KY	69
	Omaha, NE	68
	Albuquerque, NM	59
	Boston, MA	51
	Dallas, TX	50
	Miami, FL	49
	Denver, CO	46
	Chicago, IL	45
	Los Angeles, CA	42
	Seattle, WA	37
	Phoenix, AZ	35

Insight: "Atlanta reports the highest number of orders with efficient delivery times, serving as a model for operational success."

11. Payment Method Usage:

Input:

```
1 • SELECT `Payment Method`, COUNT(`Order ID`) AS Total_Orders
2 FROM pizza_sales
3 GROUP BY `Payment Method`;
```

Output:

	Payment Method	Total_Orders
▶	Card	276
	Wallet	208
	UPI	271
	Cash	202
	Domino's Cash	23
	Hut Points	24

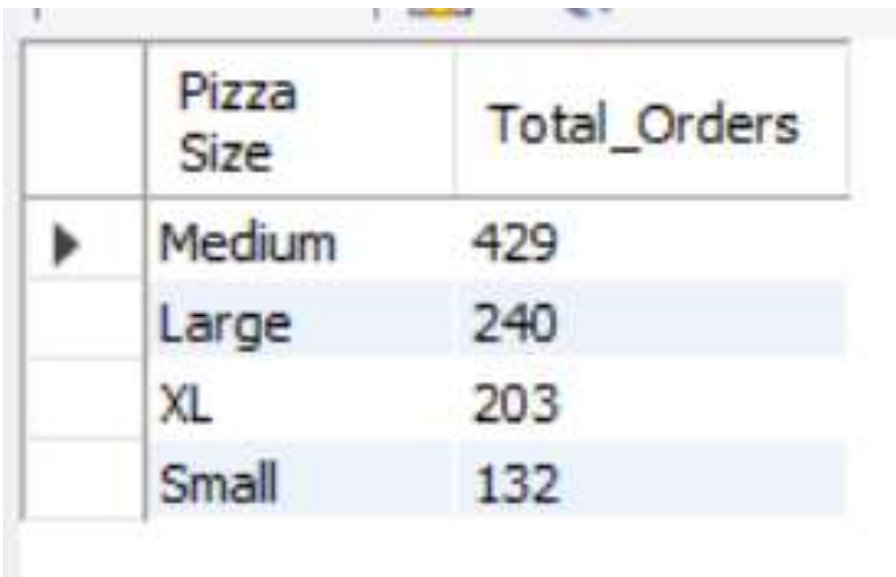
Insight: "Customers show a stronger preference for card payments compared to cash transactions."

12. Pizza Size Distribution:

Input:

```
1 • SELECT `Pizza Size`, COUNT(`Order ID`) AS Total_Orders
2 FROM pizza_sales
3 GROUP BY `Pizza Size`
4 ORDER BY Total_Orders DESC;
```

Output:

A screenshot of a database query result displayed in a table. The table has two columns: 'Pizza Size' and 'Total_Orders'. The data is sorted in descending order of 'Total_Orders'. The rows are: Medium (429), Large (240), XL (203), and Small (132). The first row is highlighted with a blue background.

	Pizza Size	Total_Orders
▶	Medium	429
	Large	240
	XL	203
	Small	132

Insight: "Medium-sized pizzas are the most frequently ordered size, identifying them as the most popular choice among customers."

1. Key Learnings:

- **Data Exploration & Schema Understanding:** Learned how to inspect table structures (DESCRIBE) and identify relevant columns for analysis.
- **SQL Proficiency:** Gained practical experience in using aggregate functions (SUM, COUNT, AVG), grouping data (GROUP BY), and sorting results (ORDER BY) to extract meaningful insights.
- **Trend Identification:** Learned how to translate raw data into business intelligence, such as identifying peak hours and correlating traffic levels with delivery delays.
- **Data-Driven Decision Making:** Realized how SQL-based insights can help optimize operations (e.g., managing inventory based on topping popularity or staffing based on peak hour trends).

2. Challenges Faced:

- **Column Identification:** Initially struggled to identify the correct column names for the analysis, which required careful inspection of the dataset and schema.
- **Data Formatting/Interpretation:** Understanding the nuances of the dataset—like why all orders showed a consistent 21-minute delay—required extra critical thinking to distinguish between actual data and potential system limitations.
- **Query Optimization:** Learning to structure queries correctly to handle datasets with special characters (using backticks ` for column names with spaces) was a key technical hurdle.
- **Synthesizing Insights:** Translating technical SQL results into simple, one-line business insights required bridging the gap between raw data and actionable recommendations.